

Lab 3: Layered Architecture Implementation (CRUD)

System: BadNet - Badminton Social Network

1. Project Structure

The following directory structure has been implemented in the BadNet project:

Bash

```
src/                # Laravel Project Root
├── app/
│   ├── Http/
│   │   └── Controllers/
│   │       ├── ChallengeController.php    # Presentation Layer
│   │       └── Services/
│   │           ├── ChallengeService.php    # Business Logic Layer
│   │           └── Repositories/
│   │               ├── ChallengeRepository.php    # Persistence Layer
│   │               └── Models/
│   │                   └── Challenge.php        # Domain Model
│   └── routes/
│       └── web.php
└── database/migrations/
    └── *_create_challenges_table.php
```

2. Implementation Details

Feature Implemented: Challenge Management (Create and Manage Challenges) — a core feature of BadNet.

Business Logic Layer: Domain Model

File: app/Models/Challenge.php

PHP

<?php

```
namespace App\Models;
```

```
use Illuminate\Database\Eloquent\Factories\HasFactory;
```

```
use Illuminate\Database\Eloquent\Model;
```

```
class Challenge extends Model
```

```
{
```

```
    use HasFactory;
```

```
    protected $fillable = [
```

```
        'challenger_id',
```

```
        'opponent_id',
```

```
        'match_date',
```

```
        'status',
```

```
        'score_challenger',
```

```
        'score_opponent',
```

```
        'elo_change_challenger',
```

```
        'elo_change_opponent'
```

```
    ];
```

```
    public function challenger()
```

```
{  
    return $this->belongsTo(User::class, 'challenger_id');  
}  
  
public function opponent()  
{  
    return $this->belongsTo(User::class, 'opponent_id');  
}  
}
```

Persistence Layer: Repository

File: app/Repositories/ChallengeRepository.php

PHP

<?php

```
namespace App\Repositories;
```

```
use App\Models\Challenge;
```

```
class ChallengeRepository
```

```
{  
    public function create(array $data)  
    {  
        return Challenge::create($data);  
    }  
}
```

```
public function findById(int $id)
{
    return Challenge::with(['challenger', 'opponent'])->find($id);
}
```

```
public function findAll()
{
    return Challenge::with(['challenger', 'opponent'])->latest()->get();
}
```

```
public function update(int $id, array $data)
{
    $challenge = Challenge::find($id);
    if ($challenge) {
        $challenge->update($data);
        return $challenge->fresh();
    }
    return null;
}
```

```
public function delete(int $id)
{
    return Challenge::destroy($id);
}
}
```

Business Logic Layer: Service

File: app/Services/ChallengeService.php

PHP

<?php

```
namespace App\Services;
```

```
use App\Repositories\ChallengeRepository;
```

```
use Illuminate\Support\Facades\DB;
```

```
class ChallengeService
```

```
{
```

```
    protected $repository;
```

```
    public function __construct(ChallengeRepository $repository)
```

```
    {
```

```
        $this->repository = $repository;
```

```
    }
```

```
    public function createChallenge(array $data)
```

```
    {
```

```
        // Business Rules
```

```
        if ($data['challenger_id'] == $data['opponent_id']) {
```

```
            throw new \Exception('Cannot challenge yourself.');
```

```
        }
```

```
if (empty($data['match_date'])) {  
    throw new \Exception('Please select a match date.');
```



```
}  
  
DB::beginTransaction();  
  
try {  
    $challenge = $this->repository->create($data);  
    DB::commit();  
    return $challenge;  
} catch (\Exception $e) {  
    DB::rollBack();  
    throw $e;  
}  
}
```

```
public function getAllChallenges()  
{  
    return $this->repository->findAll();  
}
```

```
public function getChallenge(int $id)  
{  
    $challenge = $this->repository->findById($id);  
    if (!$challenge) {  
        throw new \Exception('Challenge not found.');
```



```
}
```

```
        return $challenge;
    }
}
```

Presentation Layer: Controller

File: app/Http/Controllers/ChallengeController.php

PHP

<?php

```
namespace App\Http\Controllers;
```

```
use App\Services\ChallengeService;
```

```
use Illuminate\Http\Request;
```

```
class ChallengeController extends Controller
```

```
{
```

```
    protected $challengeService;
```

```
    public function __construct(ChallengeService $challengeService)
```

```
    {
```

```
        $this->challengeService = $challengeService;
```

```
    }
```

```
    public function index()
```

```
    {
```

```
        $challenges = $this->challengeService->getAllChallenges();
```

```
        return view('challenges.index', compact('challenges'));
    }
}
```

```
public function store(Request $request)
{
    try{
        $challenge = $this->challengeService->createChallenge($request->all());
        return redirect()->route('challenges.index')
            ->with('success', 'Challenge created successfully!');
    } catch (\Exception $e) {
        return redirect()->back()->with('error', $e->getMessage());
    }
}
```

```
public function show($id)
{
    $challenge = $this->challengeService->getChallenge($id);
    return view('challenges.show', compact('challenge'));
}
}
```

3. How to Run

1. Run database migration: `php artisan migrate`
2. Start the Laravel server: `php artisan serve`
3. Access the challenge feature at: `http://127.0.0.1:8000/challenges`